

Word Analogies with Vector Arithmetic & Virtual Threads

ATU - Dept. of Computer Science & Applied Physics Package:
ie.atu.sw | Entry point: `java -cp ./analogies.jar ie.atu.sw.Runner`

Features

1. Load Word Embeddings

Accepts a user-supplied path to any GloVe-format embeddings file (comma-separated: word followed by 50 floats). Each line is dispatched to a virtual thread for concurrent parsing, reducing load time on large files.

2. Word Analogy Operations

Users enter expressions combining words and arithmetic operators, e.g. `irish - whiskey + vodka` or `king - man + woman`. Supported operators: `+` (add), `-` (subtract), `*` (multiply), `/` (divide). Operations are applied left-to-right across all 50 vector dimensions.

3. Similarity Search with Virtual Threads

After computing the result vector, the application searches the full embeddings map for the closest matching words. Words are grouped into batches, each processed by a virtual thread, making the search efficient across large vocabularies.

4. Three Similarity Metrics

Selectable via the menu: - **Cosine similarity** (default) - angle between vectors; range -1 to 1. - **Euclidean distance** - straight-line distance between vectors. - **Dot product** - scalar sum of element-wise products.

5. Configurable Top-N Results

The number of results displayed and written to file is user-configurable (default: 10).

6. File Output

Results are written to a user-specified file (default: `./out.txt`) as a ranked list showing each word and its similarity score.

7. Coloured Console UI

ANSI colour codes provide a readable, colour-coded menu that works on Windows, Mac, and Linux terminals.